
LAPORAN PRAKTIKUM INTERNET OF THINGS

Tugas 4 & 5



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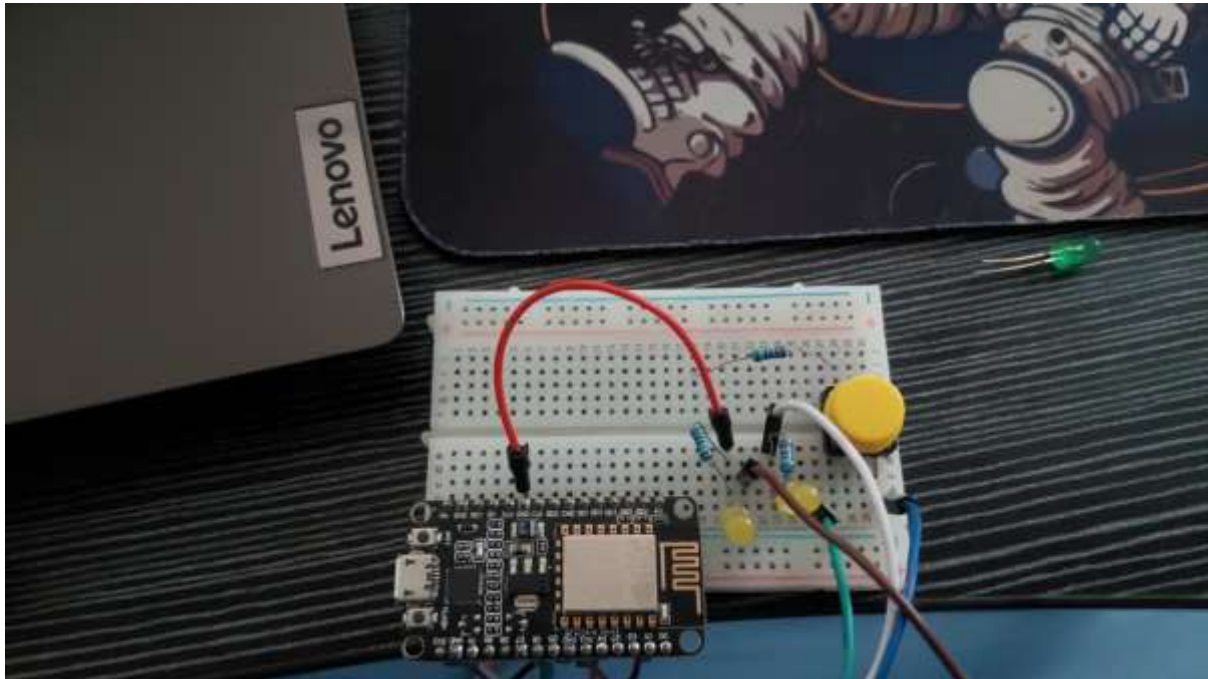
**TEKNOLOGI REKAYASA INTERNET
JURUSAN TEKNOLOGI INFORMASI
POLITEKNIK NEGERI LAMPUNG**

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4.7 Tugas: Sistem Kendali Intensitas Cahaya (PWM Slider) via WebSockets

Instruksi Perangkaian:

- Tambahkan sebuah komponen LED yang dilengkapi Resistor 220 Ohm



Instruksi Pemrograman (Front-End & Back-End):

- HTML — Menambahkan slider

```
<div class="card">
  <h2>Kontrol Intensitas LED PWM</h2>
  <input type="range" min="0" max="1023" value="0"
    id="pwmSlider" oninput="sendPWM(this.value)">
  <p>Nilai PWM: <span id="pwmValue">0</span> / 1023</p>
  <p>Intensitas: <span id="pwmPercent">0</span>%</p>
</div>
```

- JavaScript — Mengirim pesan WebSocket

```
function sendPWM(value) {
  document.getElementById('pwmValue').textContent = value;
  document.getElementById('pwmPercent').textContent =
    Math.round((Number(value) / 1023) * 100);

  if (websocket && websocket.readyState === WebSocket.OPEN) {
    websocket.send('pwm,' + value);
  }
}
```

- C++ — Mendeteksi dan memproses pesan

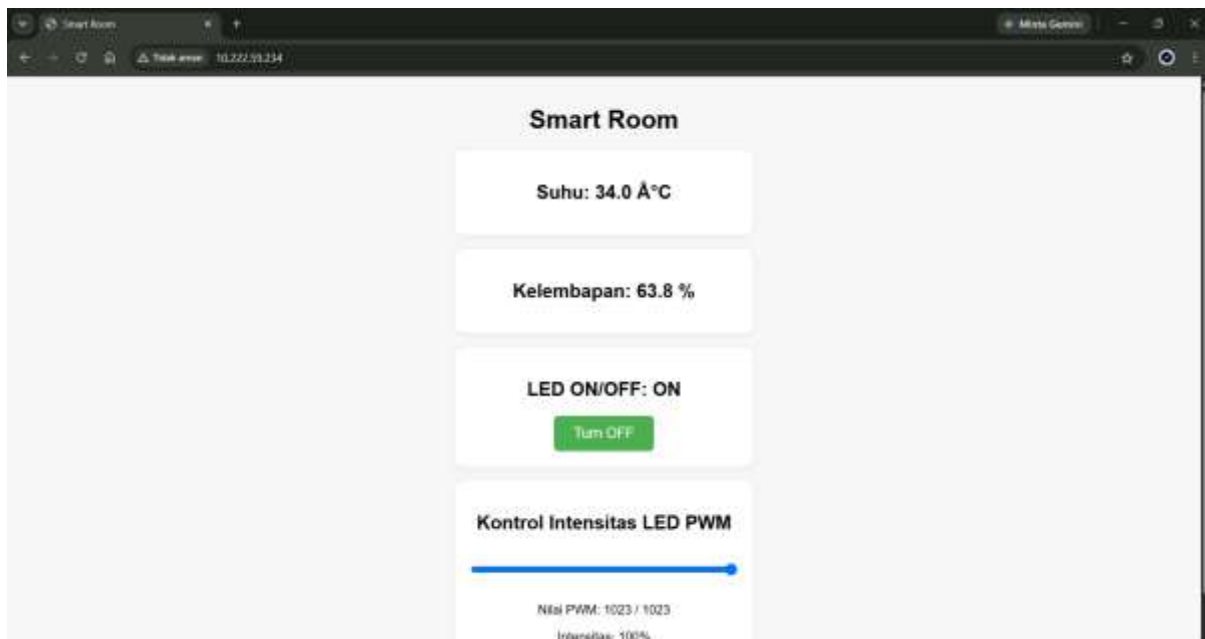
```
notifyClients();  
} else if (message.startsWith("pwm,")) {  
    int value = message.substring(4).toInt();  
    value = constrain(value, 0, 1023);  
    pwmValue = value;  
    analogWrite(pwmLedPin, pwmValue);  
    notifyClients();  
    Serial.print("PWM: ");  
    Serial.println(pwmValue);  
}
```

- Deklarasi pin dan konfigurasi PWM

```
const byte ledPin = 12;  
const byte pwmLedPin = 5;  
  
pinMode(ledPin, OUTPUT);  
pinMode(pwmLedPin, OUTPUT);  
  
digitalWrite(ledPin, LOW);  
analogWriteRange(1023);  
analogWrite(pwmLedPin, 0);
```

Kriteria Keberhasilan:

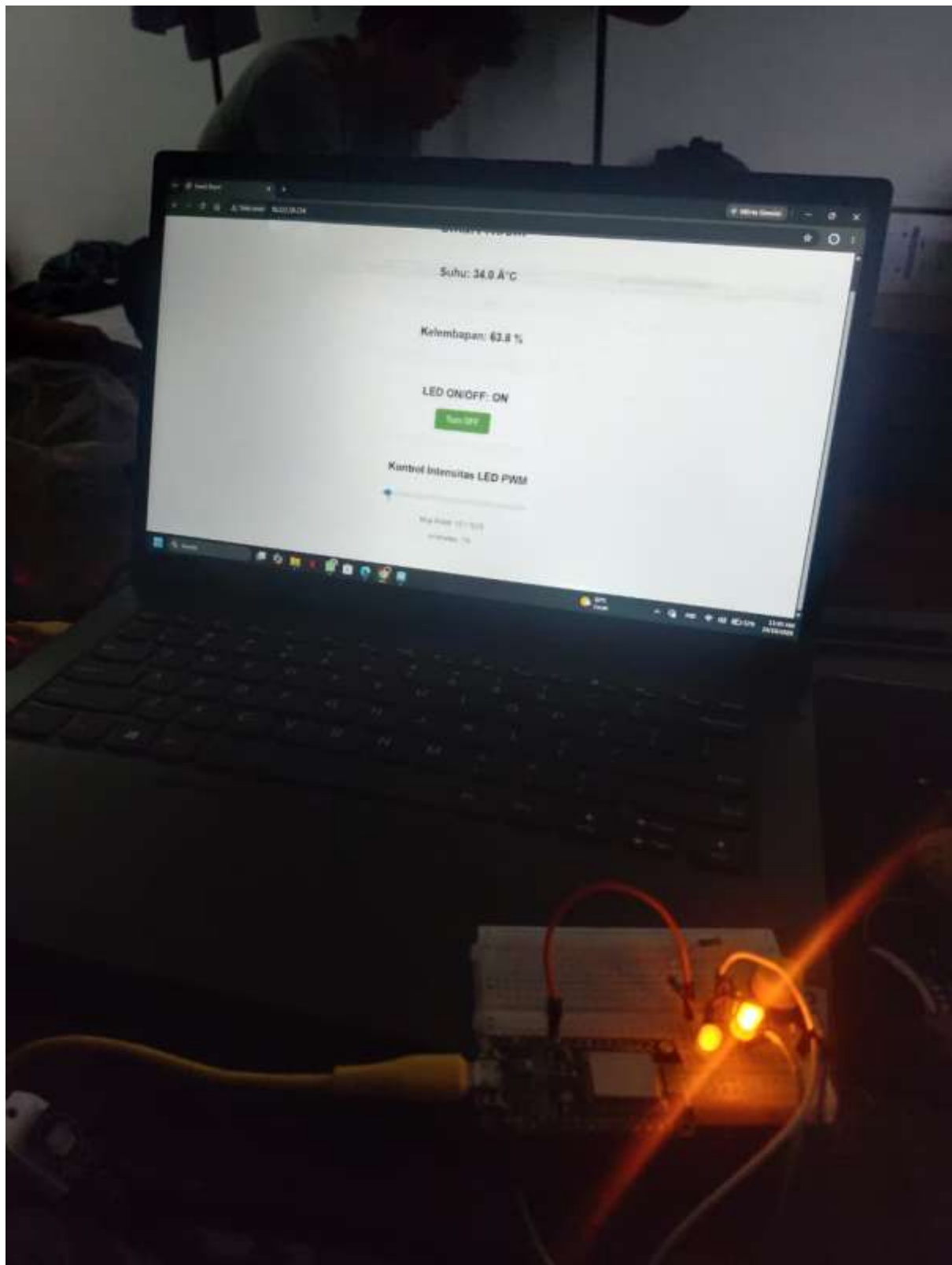
- Di layar gawai Anda, hadir tuas gulir horizontal (*Slider*)



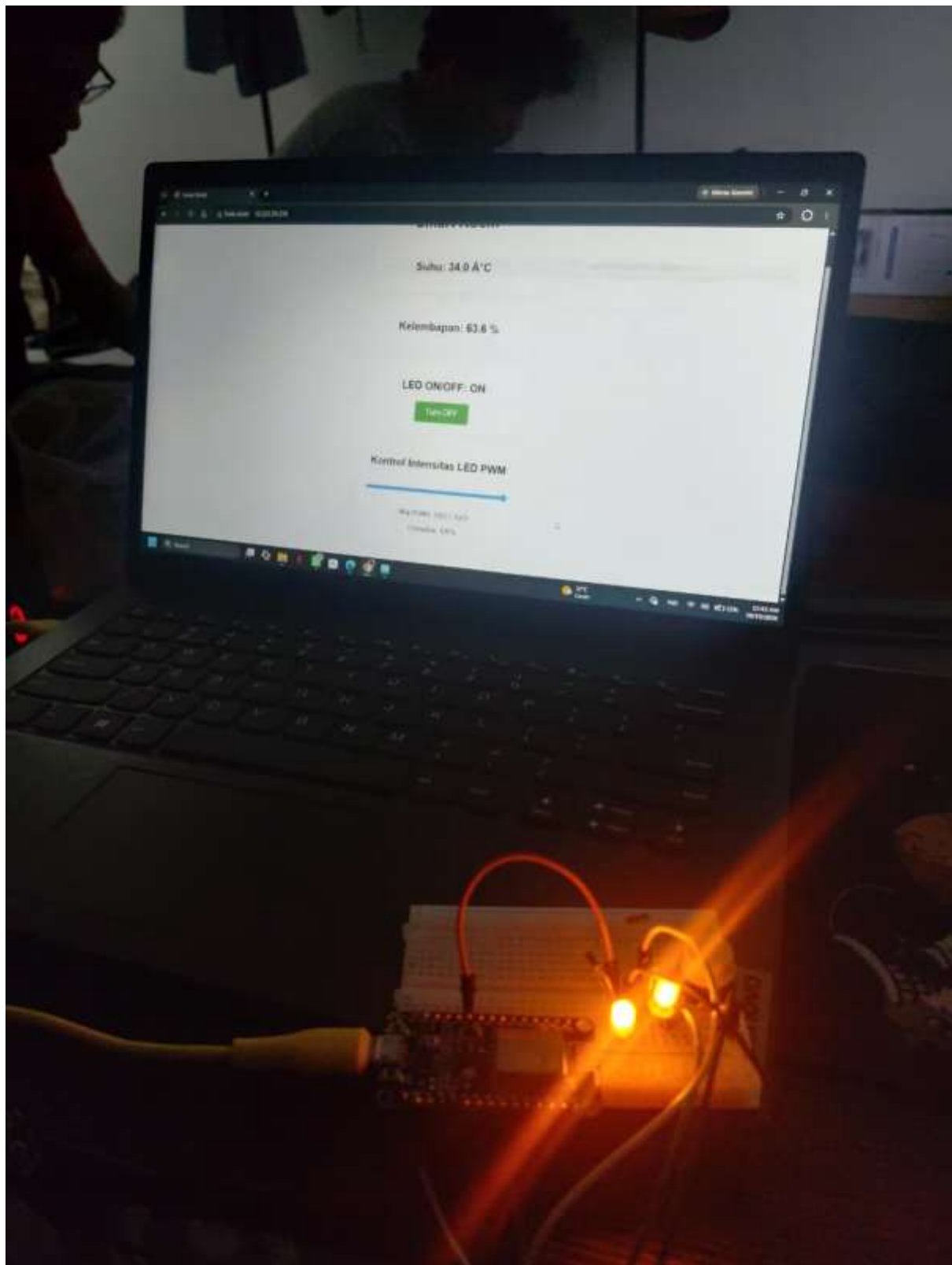
- Hasil Serial Monitor

```
11:31:10.131 -> .....
11:31:39.413 -> IP Address: 10.222.59.234
11:31:39.413 -> Suhu: 33.6 C | Kelembapan: 61.6 %
11:31:42.394 -> Suhu: 33.6 C | Kelembapan: 61.7 %
11:31:45.441 -> Suhu: 33.6 C | Kelembapan: 61.6 %
11:31:48.407 -> Suhu: 33.6 C | Kelembapan: 61.7 %
11:31:51.472 -> Suhu: 33.5 C | Kelembapan: 61.7 %
11:31:54.420 -> Suhu: 33.6 C | Kelembapan: 61.7 %
11:31:57.453 -> Suhu: 33.6 C | Kelembapan: 61.7 %
11:32:00.432 -> Suhu: 33.6 C | Kelembapan: 61.7 %
11:32:03.478 -> Suhu: 33.6 C | Kelembapan: 61.8 %
11:32:06.475 -> Suhu: 33.6 C | Kelembapan: 61.8 %
11:32:09.479 -> Suhu: 33.6 C | Kelembapan: 61.7 %
11:32:12.478 -> Suhu: 33.6 C | Kelembapan: 61.7 %
11:32:15.506 -> Suhu: 33.6 C | Kelembapan: 61.7 %
11:32:18.470 -> Suhu: 33.6 C | Kelembapan: 61.8 %
11:32:21.509 -> Suhu: 33.6 C | Kelembapan: 61.9 %
11:32:22.245 -> Client WebSocket #1 terhubung
11:32:24.529 -> Suhu: 33.6 C | Kelembapan: 61.8 %
11:32:26.649 -> PWM: 220
11:32:26.778 -> PWM: 232
11:32:26.816 -> PWM: 255
11:32:26.916 -> PWM: 268
11:32:26.951 -> PWM: 290
11:32:26.951 -> PWM: 301
11:32:26.951 -> PWM: 306
```

- Ketika Intensitas LED PWM rendah

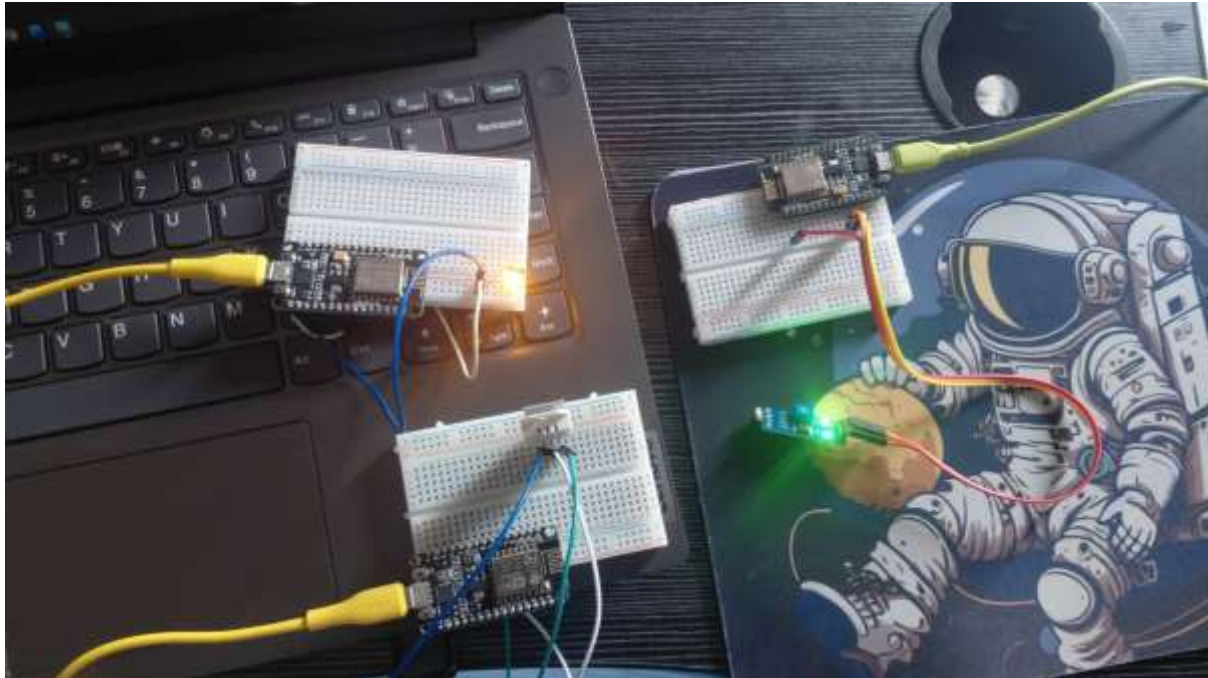


- Ketika Intensitas LED PWM Tinggi



5.7 Tugas: Sistem Telemetri & Otomasi Terdistribusi Berbasis painlessMesh

Instruksi Perangkaian



- Node 1 Sensor DHT
- Node 2: Sensor LDR
- Node 3: Hub Kontrol Aktuator

Kriteria Keberhasilan

- Hasil serial monitor Node 1 Sensor DHT

```
10 #define DHTPIN 2 // pin = GND
11
12 DHT dht(DHTPIN, DHTTYPE);
13
Output: Serial Monitor X
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM7')
12:57:06.338 -> [TK] {"tipe":"suhu_node","suhu":33.9,"kelembapan":61.7}
12:57:06.571 -> [RK] dari 3667373983: {"tipe":"cahaya_node","adc":534}
12:57:09.320 -> [TK] {"tipe":"suhu_node","suhu":33.9,"kelembapan":61.8}
12:57:09.538 -> [RK] dari 3667373983: {"tipe":"cahaya_node","adc":521}
12:57:12.299 -> [TK] {"tipe":"suhu_node","suhu":33.9,"kelembapan":61.7}
12:57:12.546 -> [RK] dari 3667373983: {"tipe":"cahaya_node","adc":600}
12:57:15.321 -> [TK] {"tipe":"suhu_node","suhu":33.9,"kelembapan":61.9}
12:57:15.540 -> [RK] dari 3667373983: {"tipe":"cahaya_node","adc":590}
12:57:18.318 -> [TK] {"tipe":"suhu_node","suhu":33.9,"kelembapan":61.7}
12:57:18.581 -> [RK] dari 3667373983: {"tipe":"cahaya_node","adc":725}
12:57:21.322 -> [TK] {"tipe":"suhu_node","suhu":33.9,"kelembapan":61.8}
12:57:21.578 -> [RK] dari 3667373983: {"tipe":"cahaya_node","adc":703}
12:57:24.296 -> [TK] {"tipe":"suhu_node","suhu":33.9,"kelembapan":61.8}
12:57:24.541 -> [RK] dari 3667373983: {"tipe":"cahaya_node","adc":727}
12:57:27.297 -> [TK] {"tipe":"suhu_node","suhu":33.9,"kelembapan":61.9}
12:57:27.571 -> [RK] dari 3667373983: {"tipe":"cahaya_node","adc":726}
```

- Hasil serial monitor Node 2 Sensor LDR

```
16  
17 // Kirim sebagai String langsung dari buffer  
18 mesh.sendBroadcast(String(msg));  
19 Serial.printf("Kirim: %s\n", msg);  
20 }  
21  
Serial Monitor X  
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM6')  
13:06:55.150 -> ~ld  
13:06:55.215 -> Kirim: {"tipe":"cahaya_node","adc":662}  
13:07:02.196 -> Kirim: {"tipe":"cahaya_node","adc":480}  
13:07:05.212 -> Kirim: {"tipe":"cahaya_node","adc":487}  
13:07:08.233 -> Kirim: {"tipe":"cahaya_node","adc":487}  
13:07:11.216 -> Kirim: {"tipe":"cahaya_node","adc":483}  
13:07:14.196 -> Kirim: {"tipe":"cahaya_node","adc":484}  
13:07:17.197 -> Kirim: {"tipe":"cahaya_node","adc":471}  
13:07:20.194 -> Kirim: {"tipe":"cahaya_node","adc":470}  
13:07:23.223 -> Kirim: {"tipe":"cahaya_node","adc":474}
```

- Hasil serial monitor Node 2 Sensor LDR

```
40 const char* tipe = doc["tipe"];
41 if (tipe == nullptr) return;
42
43 if (strcmp(tipe, "suhu_node") == 0) {
44     suhuTerakhir = doc["suhu"];
45     float hum     = doc["kelembapan"];
46     Serial.printf("[NODE 1] Suhu: %.1f C | Kelembapan: %.1f %%\n", suhuTerakhir, hum);
47 }
48 else if (strcmp(tipe, "cahaya_node") == 0) {
```